Answer key

Quick check first (⚠ = missed twice before). Full explanations start on the next page, so Warren can read why for each one he missed.

Day 1		
1	**A**	
2	**C**	⚠
3	**D**	⚠
4	**C**	
5	**C**	
6	**B**	
7	**E**	
8	**B**	
9	**E**	
10	**B**	
11	**D**	
12	**C**	
13	**D**	⚠
14	**C**	
15	**E**	
16	**C**	⚠
17	**C**	
18	**C**	⚠
19	**D**	
20	**D**	

Day 2		
21	**C**	
22	**D**	
23	**D**	
24	**B**	
25	**B**	
26	**A**	⚠
27	**A**	
28	**C**	
29	**B**	
30	**E**	⚠
31	**B**	
32	**D**	
33	**E**	⚠
34	**B**	⚠
35	**B**	
36	**B**	
37	**D**	
38	**D**	⚠
39	**D**	
40	**B**	

Day 3		
41	**A**	
42	**B**	⚠
43	**B**	⚠
44	**B**	
45	**E**	
46	**C**	
47	**B**	
48	**C**	
49	**B**	
50	**A**	
51	**C**	⚠
52	**C**	
53	**D**	
54	**B**	
55	**D**	⚠
56	**C**	
57	**A**	
58	**B**	
59	**A**	
60	**C**	

Day 4		
61	**D**	
62	**E**	
63	**E**	
64	**E**	
65	**E**	⚠
66	**C**	
67	**C**	
68	**E**	
69	**D**	
70	**A**	
71	**B**	
72	**C**	
73	**C**	
74	**A**	
75	**D**	
76	**E**	
77	**A**	⚠
78	**C**	
79	**C**	
80	**B**	

Day 5		
81	**D**	
82	**D**	
83	**B**	
84	**C**	
85	**E**	
86	**E**	
87	**A**	⚠
88	**A**	
89	**E**	
90	**B**	
91	**B**	
92	**D**	
93	**C**	
94	**D**	
95	**B**	
96	**A**	⚠
97	**E**	⚠
98	**C**	
99	**D**	

Day 6		
100	**C**	
101	**C**	
102	**C**	
103	**D**	
104	**D**	⚠
105	**D**	
106	**A**	⚠
107	**C**	
108	**D**	
109	**C**	
110	**A**	
111	**B**	
112	**E**	
113	**B**	⚠
114	**C**	
115	**D**	
116	**D**	
117	**E**	⚠
118	**A**	⚠

Day 7		
119	**E**	
120	**C**	
121	**E**	
122	**D**	
123	**C**	
124	**C**	
125	**D**	
126	**B**	
127	**C**	
128	**E**	
129	**B**	
130	**A**	
131	**E**	⚠
132	**B**	⚠
133	**E**	
134	**A**	
135	**D**	
136	**A**	
137	**C**	

Day 1 · Questions 1–20

1. **(A) 25 %** Mock 9.25b · Quant 1 · Q11

If a harvest yields 60 bushels of corn, 100 bushels of wheat, and 80 bushels of soybeans, what percent of the total harvest is corn?

Find the total first: 60 + 100 + 80 = 240. Corn is 60 out of 240 = $\frac{1}{4}$ = **25%**. (D) 40% is the trap of comparing corn to wheat only (60 out of 150). Rule: "percent of the total" — always add up the total first.

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2. **(C) 20** ⚠ MISSED TWICE Mock 9.25b · Quant 2 · Q14

A sportswriter claims that her football predictions are accurate 40% of the time. During football season, a fan kept records and found that the writer was inaccurate for a total of 30 games, although she did maintain her 40% accuracy. For how many games was the sports writer accurate?

If she is right 40% of the time, she is wrong 60% of the time. Those 30 wrong games are 60% of all games, so 10% is 5 games and the season was 50 games. Right: 40% of 50 = **20**. Check: 20 + 30 = 50, and 20 ÷ 50 = 40% ✓. (D) 40 and (E) 60 are the traps: they are the *percentages*, not numbers of games. And remember the 30 games are only the wrong ones, not all of them.

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3. **(D) $ 112.00** ⚠ MISSED TWICE Mock 9.25c · Quant 1 · Q21

Dinner (plus tax and tip) cost $ 145.6. The tax rate is 10 % and Mr. Simmons left a 20 % tip. Both tax and tip are calculated on the base amount of the check. What was the base amount of Mr. Simmons's bill?

Both extras are taken from the base, so the total is base + 10% + 20% = **130%** of the base. 145.6 ÷ 1.3 = **$112**. Check: 112 + 11.20 + 22.40 = 145.60 ✓. The slip to avoid is taking 30% off the total (that gives $101.92, not even a choice) — the 30% was worked out on the *smaller* base, not on $145.60.

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4. **(C) 45 %** Mock 9.25c · Quant 1 · Q6

The Price of a jacket is reduced by half, and the resulting price is then reduced by 10%. The final price is what percentage of the original price?

Try a price of $100. Half off: $50. Then 10% off *that*: 10% of 50 is 5, so $45. $45 out of $100 = **45%**. (B) 40% is the trap of taking 50% and 10% off the original together. Rule: when discounts come one after another, the second one is taken from the **new** price — and $100 makes the maths easy.

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5. **(C) 25 %** Mock 9.25c · Quant 2 · Q18

How much larger than 80 is 100 _____.

The increase is 20. Compare it with the number you **start from**, 80: $\frac{20}{80}$ = $\frac{1}{4}$ = **25%**. (B) 20% is the trap of comparing with 100 instead. Rule: "larger than 80" — the number after "than" goes on the bottom.

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6. **(B) 25%** Mock 10.4b · Quant 1 · Q14

If Harry eats 4 pieces of the pizza represented below, what percent of the pizza does he consume?

Count the slices: 8 lines through the center make 16 equal slices. 4 out of 16 = $\frac{1}{4}$ = **25%**. The trap is (A) 4%, which treats "4 pieces" as 4 percent. Rule: a percent is part ÷ whole, so count the whole first.

7. **(E) $\frac{3}{8}$** Mock 10.5 · Quant 1 · Q15

Of the following fractions, which is closest to 37%?

Change each to a percent: $\frac{1}{3} \approx 33.3\%$, $\frac{1}{4} = 25\%$, $\frac{2}{5} = 40\%$, $\frac{3}{7} \approx 42.9\%$, $\frac{3}{8} = 37.5\%$. The closest is **$\frac{3}{8}$**, only 0.5% away. The trap is $\frac{1}{3}$, the "famous" fraction near 37%, but it is almost 4% away. Rule: learn the eighths — $\frac{1}{8} = 12.5\%$, so $\frac{3}{8} = 37.5\%$.

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8. **(B) 10%** Mock 10.5c · Quant 2 · Q21

Chaim has $4,500 in a savings account. If last year he had $5,000 in the savings account, by what percent did the money in the account decrease from last year to now?

It went down $500. Percent decrease = 500 ÷ 5,000 (the starting amount) = **10%**. The trap is 11%, from dividing by the new amount: 500 ÷ 4,500 ≈ 11%. Rule: percent change is always measured from the original value.

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9. **(E) 0.75 m** Mock 7 · Quant 1 · Q18

Sophie is making bows for packages and uses 1.25 m of ribbon per bow. If she has 12 m and makes as many bows as possible, how much ribbon will she have left?

12 ÷ 1.25 = 9.6, and she cannot make 0.6 of a bow — so she makes **9 bows**. Those use 9 × 1.25 = 11.25 m, leaving 12 − 11.25 = **0.75 m**. The whole question is about *throwing away the decimal part* of the answer and then finding the leftover.

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10. **(B) $\frac{11}{15}$** Mock 9 · Quant 1 · Q4

What is the value of the sum of $\frac{1}{3}$ and $\frac{2}{5}$?

Fifteen is the smallest number both 3 and 5 divide into, so rewrite both fractions over 15: $\frac{1}{3} = \frac{5}{15}$ and $\frac{2}{5} = \frac{6}{15}$. Then 5 + 6 = 11 fifteenths, or **$\frac{11}{15}$**. Check it roughly: 0.33 + 0.4 = 0.73, and 11 ÷ 15 = 0.73 ✓. The trap is (A) $\frac{3}{8}$ — adding the tops and the bottoms separately. Tops add, bottoms don't.

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11. **(D) 80%** Mock 9 · Quant 1 · Q6

How will $\frac{4}{5}$ be written as a percent?

A percent is just hundredths. Fifths go into hundredths neatly: multiply top and bottom by 20 and $\frac{4}{5} = \frac{80}{100}$ = **80%**. Or divide: 4 ÷ 5 = 0.8 = 80%. (B) 125% is $\frac{5}{4}$ — the fraction flipped. Worth memorising, because they come up constantly: $\frac{1}{5} = 20\%$, $\frac{1}{4} = 25\%$, $\frac{1}{3} \approx 33\%$, $\frac{1}{8} = 12.5\%$.

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12. **(C) $\frac{2}{3}$, $\frac{4}{5}$, 0.85, $\frac{9}{10}$, $\frac{91}{100}$** Mock 9 · Quant 2 · Q23

Arrange the following numbers from least to greatest value: 0.85, $\frac{4}{5}$, $\frac{2}{3}$, $\frac{9}{10}$, $\frac{91}{100}$

Put everything in one form — decimals are easiest here. $\frac{2}{3} = 0.667$, $\frac{4}{5} = 0.8$, 0.85 stays 0.85, $\frac{9}{10} = 0.9$, $\frac{91}{100} = 0.91$. In order: **0.667, 0.8, 0.85, 0.9, 0.91** — which is choice **(C)**. (E) has the same first two but then puts 0.9 before 0.85, and (D) swaps the last pair. Two habits save you here: convert *everything* to one form before comparing anything, and when several choices start the same way, go straight to the place where they first differ.

13. (D) 28 ⚠ MISSED TWICE

Mock 9.25 · Quant 1 · Q15

$12\frac{1}{2} \div \frac{1}{2} + \frac{1}{2} \times 12 - 3 =$

Order of operations: multiply and divide first, left to right, then add and subtract. 12½ ÷ ½ = 25 (how many halves in 12½?). ½ × 12 = 6. Then 25 + 6 − 3 = **28**. Rule: × and ÷ before + and −, every time.

14. (C) 12

Mock 9.25 · Quant 1 · Q4

Evaluate: $\frac{1\frac{3}{4} - \frac{1}{4}}{\frac{1}{8}}$ _____.

Do the top first: $1\frac{3}{4} - \frac{1}{4} = 1\frac{2}{4} = 1\frac{1}{2} = \frac{3}{2}$. Now divide by $\frac{1}{8}$, which means multiply by 8: $\frac{3}{2} \times 8 =$ **12**. (B) 2 is what you get from ¼ ÷ ⅛ alone — the fraction bar means finish the whole top first. Rule: dividing by a fraction smaller than 1 makes the number *bigger*.

15. (E) $\frac{11}{32}$

Mock 9.25b · Quant 1 · Q25

On Monday, Gerri ate $\frac{1}{8}$ of an apple pie. On Tuesday, she ate $\frac{1}{4}$ of what was left of the pie. What fraction of the entire pie did Gerri eat on both days?

After Monday, $\frac{7}{8}$ of the pie is left. Tuesday she eats $\frac{1}{4}$ *of that*: $\frac{1}{4} \times \frac{7}{8} = \frac{7}{32}$. Both days: $\frac{1}{8} + \frac{7}{32} = \frac{4}{32} + \frac{7}{32} = \mathbf{\frac{11}{32}}$. (A) just repeats the $\frac{1}{4}$ from the question, but she ate $\frac{1}{4}$ of the *leftover*, and you still have to add Monday. Rule: **"of what was left" means multiply by the leftover, not by the whole.**

16. (C) $\frac{15}{46}$ ⚠ MISSED TWICE

Mock 9.25b · Quant 2 · Q7

Which of the following fractions is not more than one third?

Compare each fraction with $\frac{1}{3}$ by multiplying the top by 3. If 3 × top is bigger than the bottom, the fraction is more than a third. 66 > 63, 12 > 11, **45 < 46**, 99 > 98, 306 > 303. Only $\frac{15}{46}$ is not more than a third. (E) $\frac{102}{303}$ is the closest trap — it is only just over. Rule: **to compare with $\frac{1}{3}$, triple the top and look at the bottom** — no long division needed.

17. (C) $-\frac{2}{x}$

Mock 9.25c · Quant 1 · Q15

If $\frac{5}{x}$ is subtracted from $\frac{3}{x}$, the result is _____.

"*A* subtracted from *B*" means *B* − *A*. So it is $\frac{3}{x} - \frac{5}{x}$. Same bottom, so subtract the tops: 3 − 5 = −2, giving $\mathbf{-\frac{2}{x}}$. (D) is the trap of doing it the wrong way round (5 − 3). Rule: **"subtracted from" flips the order** — the number after "from" comes first.

18. (C) $\frac{7}{10}$ ⚠ MISSED TWICE

Mock 9.25c · Quant 1 · Q23

Which fraction lies between $\frac{2}{3}$ and $\frac{4}{5}$?

Turn everything into decimals. The window is from $\frac{2}{3} \approx 0.667$ to $\frac{4}{5} = 0.8$. The choices: 0.833, 0.85, **0.7**, 0.867, 0.9. Only $\frac{7}{10}$ fits inside. (A) $\frac{5}{6}$ is the trap — it looks like it should be in the middle, but it is bigger than 0.8.

19. (D) 120

Mock 10.5 · Quant 1 · Q23

On Monday, Aidan reads $\frac{1}{3}$ of a book, and on Tuesday, Aidan reads $\frac{1}{4}$ of the remaining pages. To finish the book, he must read an additional 60 pages. How many pages are in the book?

After Monday, $\frac{2}{3}$ is left. On Tuesday he reads $\frac{1}{4}$ of that, so $\frac{3}{4}$ of it is still left: $\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ of the book = 60 pages. The book has **120** pages. Check: 40 on Monday, 20 of the remaining 80 on Tuesday, 60 left. The trap is 144, from taking $\frac{1}{4}$ of the *whole* book. Rule: "of the remaining" means a fraction of what is left.

20. (D) $\frac{1}{9}$

Mock 10.5 · Quant 2 · Q24

What is the value of $\frac{2}{3} \times \frac{1}{8} \div \frac{3}{4}$?

$\frac{2}{3} \times \frac{1}{8} = \frac{2}{24} = \frac{1}{12}$. Dividing by $\frac{3}{4}$ means multiplying by $\frac{4}{3}$: $\frac{1}{12} \times \frac{4}{3} = \frac{4}{36} = \mathbf{\frac{1}{9}}$. The trap is $\frac{1}{16}$, from multiplying by $\frac{3}{4}$ instead of dividing. Rule: to divide by a fraction, flip it and multiply.

Day 2 · Questions 21–40

21. (C) 80 min

Mock 9 · Quant 1 · Q7

If Danny takes 48 minutes to walk 3 miles, how long should it take him to walk 5 miles maintaining the same speed?

Find the cost of *one* mile first: 48 ÷ 3 = **16 minutes per mile**. Five miles is then 5 × 16 = **80 minutes**. Sanity check: 5 miles is a bit under twice 3 miles, and 80 is a bit under twice 48 ✓. (E) 128 min comes from multiplying 48 by 8⁄3 instead of 5⁄3, and (D) 96 min is simply double. The unit rate is almost always the fastest road through a rate question — get to "per one" and the rest is one multiplication.

22. (D) 27 nights

Mock 9 · Quant 2 · Q10

If Sarah reads at an average rate of 21 pages in four nights, how long will it take her to read 140 pages?

Get to one night: 21 ÷ 4 = **5.25 pages a night**. Then 140 ÷ 5.25 = **26.67** nights — and since she cannot stop two-thirds of the way through a night, she needs a **27th** night to finish. (B) 26 nights leaves her 3½ pages short; (A) 6 is 140 ÷ 21, which forgets that each of those 21-page blocks takes four nights. When time comes out as a decimal, ask what happens in the leftover bit — finishing a job almost always rounds *up*.

23. (D) 390

Mock 9 · Quant 2 · Q19

In Jim's school, there are 3 girls for every 2 boys. There are 650 students in total. Using this information, how many students are girls?

A ratio question is a sharing question. Three girls and two boys make a group of **5 parts**, so one part is 650 ÷ 5 = **130** students. Girls take 3 parts: 3 × 130 = **390**. Check: boys are 2 × 130 = 260, and 390 + 260 = 650 ✓. (A) 260 is the number of boys and (B) 130 is one part — both are real steps on the way, which is exactly why they are in the list. Find one part first, then take the shares the question asks for.

24. (B) 50% Mock 9 · Quant 2 · Q24

Keith's bakery had 252 customers go through its doors last week. This week, that number increased to 378. Express this increase as a percentage.

Percent increase is **the change divided by the starting number**. The change is 378 − 252 = **126**, and 126 ÷ 252 = 0.5 = **50%**. It helps to notice 126 is exactly half of 252 — no long division needed. Dividing by the *new* number instead (126 ÷ 378 ≈ 33%) is the single most common percent-change mistake there is. The starting value always goes on the bottom.

25. (B) $1\frac{5}{8}$ inches Mock 9.25 · Quant 1 · Q14

How long should an object $6\frac{1}{2}$ feet long be drawn, if according to the scale, $\frac{1}{4}$ inch in the drawing equals 1 foot?

Each real foot becomes $\frac{1}{4}$ inch. So $6\frac{1}{2}$ feet becomes $6\frac{1}{2} \times \frac{1}{4} = \frac{13}{2} \times \frac{1}{4} = \frac{13}{8}$ = **$1\frac{5}{8}$ inches**. Check: 6 feet → 1½ inches, and the extra ½ foot → $\frac{1}{8}$ inch; $1\frac{4}{8} + \frac{1}{8} = 1\frac{5}{8}$. (A) $1\frac{3}{4}$ is the trap of rounding 6½ up to 7. Rule: scale problems are one multiplication — real size × scale.

26. (A) 2 $\frac{1}{2}$ inches ⚠ MISSED TWICE Mock 9.25b · Quant 2 · Q11

A snapshot measures 1 $\frac{1}{4}$ inches by 2 inches. It is to be enlarged so that the longer dimension will be 4 inches. The length of the enlarged shorter dimension will be _____.

The longer side goes from 2 to 4 inches — it **doubles**. An enlargement keeps the shape, so the shorter side doubles too: $2 \times 1\frac{1}{4}$ = **$2\frac{1}{2}$** inches. The classic slip is to *add* 2 inches to each side (giving $3\frac{1}{4}$) — but enlarging multiplies, it never adds. Rule: enlarging a picture multiplies every length by the same number.

27. (A) 0.275 Mock 9.25b · Quant 2 · Q16

55 is to _____ as 110 is to 0.55.

55 is **half** of 110, so the blank must be half of 0.55: 0.55 ÷ 2 = **0.275**. Check: 110 ÷ 0.55 = 200 and 55 ÷ 0.275 = 200 ✓ — the same ratio both times. Rule: find how the two known numbers are linked (here "half") and do the same thing to the other side.

28. (C) 1 $\frac{1}{4}$ inches Mock 9.25c · Quant 2 · Q19

If $\frac{3}{8}$″ on a scale drawing is equivalent to one foot at full scale, what distance on the drawing will stand for forty inches?

Match units first: one foot is 12 inches. So 12 real inches → $\frac{3}{8}$ inch on paper, which means 1 real inch → $\frac{3}{8} \div 12 = \frac{1}{32}$ inch. Forty inches: $40 \times \frac{1}{32} = \frac{40}{32}$ = **$1\frac{1}{4}$** inches. Sense check: 40 inches is a bit over 3 feet, and $3 \times \frac{3}{8} = 1\frac{1}{8}$ ✓. Rule: **convert to the same unit before scaling.**

29. (B) $1\frac{5}{27}$ Mock 8 · Quant 1 · Q11

Find the value of y in the equation $y = 4x^3$ when $x = 2/3$.

Cube first, multiply second — the exponent belongs to x alone, not to $4x$. $(2/3)^3 = 8/27$, and $4 \times 8/27 = 32/27$ = **$1\frac{5}{27}$**. The trap is cubing $4x = 8/3$ to get 512/27 (about 19), far too big. Order of operations beats instinct every time.

30. **(E) $d = e$** ⚠ MISSED TWICE — Mock 8 · Quant 1 · Q12

Review the following five algebraic expressions and then select the correct answer below.

(a) $4(r + s) + t$

(b) $4r + 4s$

(c) $4rst$

(d) $4r(s + t)$

(e) $4rt + 4rs$

Expand (d): $4r(s + t) = 4rs + 4rt$, which is (e) with the two terms written in the other order — so $\boldsymbol{d = e}$, always, whatever the letters stand for. Every other choice is a comparison that depends on the values: (a) = (b) only when $t = 0$, and (B) and (D) can go either way depending on whether the letters are big, small or negative. On a question like this, look for the pair that is **identically** equal — an *equals* that survives every substitution beats a *greater than* that only sometimes holds.

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31. **(B) $\frac{3t + 2}{3}$** — Mock 8 · Quant 1 · Q20

Hal spends $t + 4$ hours per day watching television. José spends t hours and Tony watches television $t - 2$ hours per day. Find the average number of hours of watching television for the three students.

An average with letters works exactly like an average with numbers: add, then divide by how many. $(t + 4) + t + (t - 2) = 3t + 2$, and there are three students, so the mean is $\boldsymbol{(3t + 2)/3}$. The only real work is the +4 and the −2 — combine them to +2, and choice (E) with its −2 is gone. Do not try to simplify $(3t + 2)/3$ into $t + 2$; only the $3t$ divides cleanly.

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32. **(D) $y = 2x - 1$** — Mock 8 · Quant 1 · Q24

Express the linear relationship indicated in the table.

Each time x goes up by 1, y goes up by 2 — so the rule starts $2x$. Now fix the offset with any single pair: at $x = 1$, $2x = 2$ but $y = 1$, so subtract 1: $\boldsymbol{y = 2x - 1}$. Test it on the far end, where mistakes show up: $x = 5$ gives $10 - 1 = 9$. ✓ Choice (B) fits none of the pairs; always check a rule against *two* rows, ideally the first and the last.

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33. **(E) 18** ⚠ MISSED TWICE — Mock 8 · Quant 1 · Q8

Hector Villanueva has a 0.300 batting average (hits/times at bat). If he gets a total of 10 more hits the next 10 times at bat, his batting average will jump to 0.400. Determine how many hits he had originally.

Name the at-bats n. Then his hits are $0.3n$, and afterwards $(0.3n + 10)/(n + 10) = 0.4$. Cross-multiply: $0.3n + 10 = 0.4n + 4$, so $6 = 0.1n$ and $n = 60$ at-bats. The question asks for **hits**, not at-bats: $0.3 \times 60 =$ **18**. Check: 18/60 = .300, and 28/70 = .400. ✓ Always re-read the last line before you circle — the number you solved for and the number they asked for are often different.

34. **(B) 1** ⚠ MISSED TWICE — Mock 8 · Quant 1 · Q9

Given $5 > 3x \geq -2$, find a possible value for x.

Divide the whole chain by 3: $5/3 > x \geq -2/3$, that is x is under about 1.67 and at least about −0.67. Only **1** sits inside. Or skip the algebra and test the choices in $3x$: 3(2) = 6, too big; 3(1) = 3, which is under 5 and above −2 ✓; 3(−1) = −3, below −2. On a "possible value" question, substituting the choices is usually faster than solving.

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35. **(B) −10** — Mock 8 · Quant 2 · Q16

If $x = 4/5$, $y = 2/3$, and $z = \sqrt{3}$, find the value of $10x - 9y - 4z^2$.

Each term is built to cancel neatly. $10 \times 4/5 = 8$. $9 \times 2/3 = 6$. And $z^2 = (\sqrt{3})^2 = \mathbf{3}$, so $4z^2 = 12$. Put it together: $8 - 6 - 12 = \mathbf{-10}$. The point of the question is that last step: **squaring a square root undoes it**, so you never have to touch a decimal. Anyone who reaches for a value of $\sqrt{3}$ is doing work the question did not ask for.

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36. **(B) $3x^2 - 3x + 11$** — Mock 9 · Quant 1 · Q10

How will the following algebraic expression be simplified: $(5x^2 - 3x + 4) - (2x^2 - 7)$?

The minus sign in front of the bracket belongs to *everything* inside it, so $-(2x^2 - 7)$ becomes $-2x^2 + 7$. Now collect: $5x^2 - 2x^2 = 3x^2$, the $-3x$ has nothing to pair with, and 4 + 7 = 11 — giving $\mathbf{3x^2 - 3x + 11}$. (C) is exactly what you get if you forget to flip that second sign (4 − 7 = −3), which is the single most common slip in the whole topic. Write the sign change in before you collect anything.

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37. **(D) $x = -5$** — Mock 9 · Quant 1 · Q12

If $-3(x + 4) = x + 8$, what is the value of x ?

Multiply out first, keeping the negative on both terms: $-3x - 12 = x + 8$. Gather the x's on one side and the numbers on the other: $-3x - x = 8 + 12$, so $-4x = 20$ and $\boldsymbol{x = -5}$. Always substitute back: $-3(-5 + 4) = -3(-1) = 3$, and $-5 + 8 = 3$ ✓. Forgetting to distribute the minus over the 4 gives $-3x + 12$ and the wrong sign on the answer — the check catches it in five seconds.

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38. **(D) $x \geq 1$** ⚠ MISSED TWICE — Mock 9 · Quant 1 · Q14

Which inequality represents the values displayed on the number line?

Read the dot, then the arrow. The dot on 1 is **filled in**, so 1 itself is included — that means ≤ or ≥, never < or >. The heavy line runs to the **right**, towards the bigger numbers, so the values are 1 and everything above it: $\boldsymbol{x \geq 1}$. (C) $x > 1$ is the trap for anyone who reads the arrow but not the dot. Two things to check every time: is the circle hollow or solid, and which way does the ray point?

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39. **(D) $23/8$** — Mock 9 · Quant 2 · Q22

Solve this equation: $9x + x - 7 = 16 + 2x$

Tidy each side before you move anything: $9x + x = 10x$, so the equation is $10x - 7 = 16 + 2x$. Take $2x$ off both sides: $8x - 7 = 16$. Add 7: $8x = 23$. Divide: $\boldsymbol{x = 23/8}$. An ugly fraction feels wrong, but check it — $8(23/8) = 23$ and $23 - 7 = 16$ ✓. (C) $9/8$ comes from moving the 7 the wrong way ($16 - 7$ instead of $16 + 7$). Answers on the SSAT are allowed to be fractions; do not talk yourself out of one.

40. (B) 2 Mock 9 · Quant 2 · Q25

If $4x - 3 = 5$, then x = ?

Undo the operations in reverse order. The equation does "times 4, then minus 3", so you undo with "plus 3, then divide by 4": $4x = 8$, so $\boldsymbol{x = 2}$. Check: $4(2) - 3 = 5$ ✓. With five short choices you can also just test them — on a last question with the clock running, substituting (B) takes three seconds and proves it. Undo in reverse, then verify; both are fast here.

Day 3 · Questions 41–60

41. (A) larger than 7 Mock 9.25 · Quant 1 · Q16

If $y + 3 > 10$, the y may be _____.

Solve it like an equation: take 3 from both sides and $y > 7$. So y is **larger than 7**. (B) is the trap: $y = 6.5$ is larger than 6 but does not work, since $6.5 + 3 = 9.5$, not more than 10. Rule: an inequality is solved exactly like an equation (as long as you do not multiply or divide by a negative).

42. (B) $-2 \le x < 3$ ⚠ MISSED TWICE Mock 9.25 · Quant 1 · Q23

The graph shown above is of which inequality?

Look at the two ends. At −2 the dot is **filled in**, so −2 is included: $-2 \le x$. At 3 the circle is **open**, so 3 is not included: $x < 3$. Together: $\boldsymbol{-2 \le x < 3}$. (D) swaps the two ends. Rule: filled dot = ≤ or ≥ (included); open circle = < or > (not included).

43. (B) $\frac{a^2+b^2}{ab}$ ⚠ MISSED TWICE Mock 9.25 · Quant 1 · Q25

What is the sum of $\frac{a}{b}$ and $\frac{b}{a}$?

Use the common bottom ab: $\frac{a}{b} = \frac{a \cdot a}{ab}$ and $\frac{b}{a} = \frac{b \cdot b}{ab}$. Add the tops: $(a^2 + b^2)$ over ab. That is **(B)**. Check with numbers: a = 1, b = 2 gives ½ + 2 = 2½, and (1 + 4) ÷ 2 = 2½. ✓ (C) is the trap of treating $a^2 + b^2$ as $(a + b)^2$. Rule: when you are unsure with letters, try small numbers.

44. (B) $20b$ Mock 9.25 · Quant 1 · Q8

If $a = 4$ and $b = \frac{1}{5}$, then the value of a, expressed in terms if b, is _____.

Ask: how many b's make 4? Each b is $\frac{1}{5}$, and $4 \div \frac{1}{5} = 4 \times 5 = 20$. So $a = \boldsymbol{20b}$. Check: $20 \times \frac{1}{5} = 4$. ✓ (A) $25b$ is the trap of using 5×5. Rule: to write a number "in terms of b", divide it by b and put b back on.

45. (E) 2 Mock 9.25b · Quant 1 · Q20

If $|\,4a - 3\,| = 5$, which of the following is a possible value for a?

Absolute value means "distance from zero", so $4a - 3$ is either 5 or −5. If it is 5: $4a = 8$, $a = \mathbf{2}$. If it is −5: $4a = -2$, $a = -\frac{1}{2}$ (not a choice). Check (E): $|8 - 3| = 5$ ✓. Quicker still: **plug the choices in** — only 2 works.

46. **(C) $2x - 3$** Mock 9.25b · Quant 2 · Q1

Bob is x years old and Jerry is 5 years older. In terms of x, what was the sum of their ages, in years, 4 years ago?

Write each age *now*, then go back 4 years. Bob: $x \to x - 4$. Jerry: $x + 5 \to x + 1$. Sum: $(x - 4) + (x + 1) =$ **$2x - 3$**. Check with a number: if Bob is 10, Jerry is 15; four years ago they were 6 and 11, total 17, and $2(10) - 3 = 17$ ✓. (A) $2x + 3$ has a sign the wrong way: with Bob = 10 it gives 23, not 17. Rule: **test your expression with a real number.**

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47. **(B) $\frac{8x+4}{3}$** Mock 9.25b · Quant 2 · Q8

The length of each side of the square above is $\frac{2x+1}{3}$. The perimeter of the square is _____.

A square has 4 equal sides, so the perimeter is $4 \times \frac{2x+1}{3} = \mathbf{\frac{8x+4}{3}}$. The 4 multiplies the *whole* top, both the $2x$ and the 1. (A) is the trap: it multiplies the $2x$ by 4 but then adds the 4 outside the fraction instead of dividing it by 3. Check with $x = 1$: each side is 1, perimeter 4; (B) gives $12 \div 3 = 4$ ✓, (A) gives $6\frac{2}{3}$.

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48. **(C) $81x^{11}y^{-15}z^4$** Mock 9.25 · Quant 2 · Q23

$(3x^2y^{-4}z)^4x^3y =$ _____.

Raise every part of the bracket to the 4th power: $3^4 = 81$, $(x^2)^4 = x^8$, $(y^{-4})^4 = y^{-16}$, z^4. Then multiply by x^3y by adding exponents: $x^{8+3} = x^{11}$ and $y^{-16+1} = y^{-15}$. Result: **$81x^{11}y^{-15}z^4$**. (A) is the trap of forgetting to raise the 3 to the 4th power. Rule: a power outside a bracket hits *everything* inside, numbers included. (The paper prints this choice a little jumbled; we show it cleanly.)

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49. **(B) $500y$** Mock 9.25c · Quant 1 · Q10

At a fund-raiser, 500 people each donated y dollars. In terms of y, what was the total number of dollars donated?

Each of 500 people gave y dollars, so the total is 500 lots of y: **$500y$**. Test it: if everyone gave \$2, the total is \$1,000, and $500 \times 2 = 1{,}000$ ✓; (E) would give only 502. Rule: "each" usually means **multiply**.

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50. **(A) $9K^6$** Mock 9.25c · Quant 1 · Q13

The expression $(3K^3)^2$ is equivalent to _____.

Squaring means multiplying the whole thing by itself: $(3K^3)(3K^3)$. Numbers: $3 \times 3 = 9$. Powers: $K^3 \times K^3 = K^6$ (add the exponents). So **$9K^6$**. (D) $9K^5$ adds $3 + 2$ instead of multiplying 3×2; (B) cubes the 3 instead of squaring it. Rule: a power of a power **multiplies** the exponents.

---

51. **(C) $-\frac{5}{4}$** ⚠ MISSED TWICE Mock 9.25c · Quant 1 · Q18

What is the x-intercept of the line described by the equation $y = 4x + 5$?

The x-intercept is where the line crosses the x-axis, and there **$y = 0$**. So $0 = 4x + 5$, $4x = -5$, $x = \mathbf{-\frac{5}{4}}$. (A) 5 is the trap: that is the y-intercept, where $x = 0$. Rule: x-intercept → set $y = 0$; y-intercept → set $x = 0$.

52. (C) 5 Mock 9.25c · Quant 2 · Q22

If $a = 16$, $b = 2$, and $c = 3$, the value of $\sqrt{a + 3b + c}$ is _____.

Work inside the square root first: 16 + 3 × 2 + 3 = 16 + 6 + 3 = 25. √25 = **5**. (D) 4 is the trap of taking √16 and forgetting the rest.

53. (D) $(w + 5)w$ Mock 10.4 · Quant 1 · Q21

If the length of a rectangle is 5 more than the width, w, what is the area of the rectangle in terms of w?

"5 more than the width" means length = $w + 5$. Area = length × width = $\boldsymbol{(w + 5)w}$. The traps: $5w$ means 5 *times* the width, and $w - 5$ is 5 *less*. Rule: "more than" is adding; "times" is multiplying.

54. (B) $\frac{3}{7}$ Mock 10.5 · Quant 2 · Q20

If $3 + 7 \times n = 6$, what is the value of n?

Multiplication comes first, so this says $3 + 7n = 6$. Subtract 3: $7n = 3$, so $n = \frac{3}{7}$. The trap is $\frac{3}{5}$, from adding 3 + 7 first ($10n = 6$). Rule: order of operations still applies inside equations.

55. (D) –2 ⚠ MISSED TWICE Mock 8 · Quant 1 · Q7

Simplify $6|-3| - 4|5|$.

Absolute value bars first: $|-3| = 3$ and $|5| = 5$. So the expression is 6(3) – 4(5) = 18 – 20 = **–2**. The bars only strip the sign off what is *inside* them — the minus sign in the middle of the expression is a subtraction and stays exactly where it is. That is the trap in (B): treating the whole answer as positive because absolute values were involved.

56. (C) $2\frac{1}{4}$ Mock 8 · Quant 2 · Q23

Find the value of $\frac{3^9}{4 \times 3^7}$.

Cancel the powers of 3 before you calculate anything: $3^9 \div 3^7 = 3^{9-7} = 3^2 = 9$. That leaves 9/4 = **2¼**. Working out $3^9 = 19{,}683$ and $3^7 = 2{,}187$ gets there too, but it is four times the work and four times the risk. **When you divide powers of the same base, subtract the exponents** — and do it first, while the numbers are still small.

57. (A) 5 × 3 – (2 + 7) Mock 8 · Quant 2 · Q24

Of the following expressions, which is the smallest?

There is no shortcut — evaluate all five, keeping strictly to brackets, then powers, then × and ÷, then + and –. (A) 15 – 9 = **6**. (B) 15 – 8 = 7. (C) 14 – 5 + 5 = 14. (D) 9 – 3(–2) + 6 = 9 + 6 + 6 = 21. (E) 45 – 9 = 36. The smallest is **(A)**, and (B) is only one away, so this is a question you lose by rushing rather than by not knowing. Watch (D) especially: subtracting a negative makes the total *bigger*.

58. (B) 13,078 Mock 9 · Quant 1 · Q2

How will the following number be written in standard form: $(1 \times 10^4) + (3 \times 10^3) + (7 \times 10^1) + (8 \times 10^0)$?

Write the empty place in. $10^4 = 10{,}000 \cdot 10^3 = 1{,}000 \cdot 10^1 = 10 \cdot 10^0 = 1$, so the sum is 10,000 + 3,000 + 70 + 8. There is **no 10^2 term**, so the hundreds place is a **zero**: **13,078**. That missing term is the whole question — (C) 1,378 is what you get by writing the digits 1, 3, 7, 8 in a row and skipping the empty place. Whenever a place is missing from expanded form, put a zero there on purpose.

59. (A) 847.90 Mock 9 · Quant 1 · Q3

How will the number 847.89632 be written if rounded to the nearest hundredth?

The hundredths place is the second digit after the point, so you are keeping 847.**89** and the next digit decides. That digit is **6**, which is 5 or more, so the 9 rounds up: 847.89 + 0.01 = **847.90**. (C) 847.89 is the trap — it is what you get by chopping the tail off instead of rounding it. Keep the trailing zero: it is what shows you rounded to hundredths and not to tenths.

60. (C) 34 Mock 9 · Quant 1 · Q5

What is the value of the expression: $7^2 - 3 \times (4 + 2) + 15 \div 5$?

Work the order of operations in blocks. Brackets first: 4 + 2 = 6. Then the power: $7^2 = 49$. Then *all* the multiplying and dividing: 3 × 6 = 18 and 15 ÷ 5 = 3. Only now the adding and subtracting, left to right: 49 − 18 + 3 = **34**. The trap is doing 49 − 18 + 3 out of order as 49 − 21 = 28, or adding before subtracting. Plus and minus have equal rank — you take them left to right, never one before the other.

Day 4 · Questions 61–80

61. (D) 1, 2, 3, 4, 6, 12 Mock 9 · Quant 1 · Q8

What are all the factors of 12?

Factors go *into* a number; multiples come *out* of it. Hunt factors in pairs so you cannot miss one: 1 × 12, 2 × 6, 3 × 4 — and 5 does not divide 12, so you are done. That gives **1, 2, 3, 4, 6, 12**. (A) and (C) are lists of multiples; (B) has dropped the 3; (E) has dropped 1 and 12, which always count. Pairing up is the habit: it tells you when to stop.

62. (E) 1.43785 Mock 9 · Quant 2 · Q1

Which of the following numbers has the greatest value?

Compare place by place, left to right, and stop at the first difference — never by how many digits a number has. (D) starts with 0, so it is out. (B) has 0 in the tenths place, so it is out. That leaves 1.4378**0**, 1.435… and 1.4378**5**. In the thousandths place 7 beats 5, so (C) goes. The last two agree as far as 1.4378, so line them up and pad with a zero: 1.43780 against **1.43785** — (E) wins by five hundred-thousandths. The trap is thinking the longest string of digits is biggest; padding with zeros turns the comparison into one you can see.

63. (E) 2.551 Mock 9.25 · Quant 1 · Q1

$1\frac{1}{2} + 0.750 + 0.3010 =$ _____.

Turn the mixed number into a decimal first: $1\frac{1}{2} = 1.5$. Then line the decimal points up: 1.500 + 0.750 + 0.301 = **2.551**. The trap is (B) 2.051 — what you get if the ½ is dropped or the columns slip. Rule worth keeping: when decimals are added, pad them to the same number of places so every digit sits in its own column.

64. (E) 12% Mock 9.25 · Quant 1 · Q12

What percentage of 220 is 26.4?

26.4 is much smaller than 220, so the answer must be a small percent — that rules out (A), (B) and (C) straight away. 10% of 220 = 22, and 2% = 4.4, so 12% = 26.4. **12%**. (A) 909% is the trap of dividing the wrong way (220 ÷ 26.4 ≈ 8.3). Rule: part ÷ whole × 100 — and check that the answer is sensible in size.

65. (E) 5,150 ÷ 10,000 ⚠ MISSED TWICE Mock 9.25 · Quant 1 · Q19

0.0515 × 10 is equivalent to _____.

First the target: 0.0515 × 10 = 0.515 (the point moves one place right). Now test each choice. (A) = 51.5, (B) = 51.5, (C) = 5.15, (D) = 51,000, (E) 5,150 ÷ 10,000 = **0.515**. Rule: ×10 moves the point one place right; ÷10,000 moves it four places left. Count the places, do not guess.

66. (C) 9.08 Mock 9.25 · Quant 1 · Q3

565.684 ÷ 62.3 = _____.

Estimate first: 565 ÷ 62 is a bit more than 9 (62 × 9 = 558). That already rules out (A), (B) and (E). To choose between 9.08 and 9.081, check: 62.3 × 9.08 = 565.684 exactly, so the answer is **9.08**. (D) 9.081 is the trap for students who guess a longer decimal must be more exact. Rule: estimate to throw out the far-off choices, then multiply back to check the close ones.

67. (C) 3 Mock 9.25 · Quant 1 · Q6

−3 − [(4 − 3) − (3 + 4)] = _____.

Work from the inside out. (4 − 3) = 1 and (3 + 4) = 7, so the bracket is 1 − 7 = −6. Then −3 − (−6) = −3 + 6 = **3**. The trap is (E) −9, from forgetting that subtracting a negative means adding. Rule: two minus signs side by side turn into a plus.

68. (E) 60 Mock 9.25c · Quant 2 · Q10

Durant's Trading Company earned profits of $ 75,000 in 1990. In 1998, their profit was $ 4,500,000. The profit from 1998 was how many times as great as it was in 1990?

"How many times as great" means divide: 4,500,000 ÷ 75,000. Cross off three zeros from each: 4,500 ÷ 75 = **60**. Check: 75,000 × 60 = 4,500,000 ✓. (C) 6 is the trap of losing a zero.

69. (D) 6 Mock 10.4 · Quant 1 · Q19

In the multiplication problem shown, ◇ is equal to which of the following digits?

Undo the multiplication: 1,168 ÷ 8 = 146, so the hidden digit is **6**. Check: 6 × 8 = 48 ends in 8, matching the last digit of 1,168. Trying 5 gives 145 × 8 = 1,160, which is too small. Rule: the ones digit of the answer is a fast first test.

70. (A) 8.302

Mock 10.4b · Quant 1 · Q8

What is the second to greatest of the numbers below?

Compare digit by digit after the decimal point: tenths first. 8.320 and 8.302 both have 3 tenths; 8.320 is bigger (2 hundredths vs 0), so it is the greatest and **8.302** is second. The trap is (E) 8.320, which is the greatest, not the second greatest. Rule: line the decimals up and compare one place at a time from the left.

71. (B) 3

Mock 10.5 · Quant 1 · Q4

4,■86

The ■ in the number above represents a digit from 0 through 9. If the number is less than 4,486, what is the greatest possible value for ■?

Compare place by place: the thousands digits (4) match, so the hundreds digit decides. To be *less than* 4,486, the hidden digit must be smaller than 4, and the greatest such digit is **3** (4,386). The trap is 4, which makes 4,486 — equal, not less. Rule: "less than" never includes the number itself.

72. (C) 400

Mock 10.5 · Quant 2 · Q11

Of the following, which is the closest value to $\frac{998 \times 1{,}004}{48 \times 52}$?

Round: 998 × 1,004 ≈ 1,000 × 1,000 = 1,000,000, and 48 × 52 ≈ 50 × 50 = 2,500. Then 1,000,000 ÷ 2,500 = **400**. The traps 40 and 4,000 come from losing or adding a zero. Rule: round to friendly numbers and count zeros carefully.

73. (C) 2, 3, 5, 7, 11

Mock 10.5 · Quant 2 · Q13

Which of the following lists contains only prime numbers?

A prime has exactly two factors: 1 and itself. **2, 3, 5, 7, 11** are all prime. The others each hide a non-prime: 0 and 1 are not prime, 4 = 2 × 2, 15 = 3 × 5, and 9 = 3 × 3. Rule: 1 is not prime, and watch for odd numbers like 9 and 15 that look prime but are not.

74. (A) 32

Mock 8 · Quant 1 · Q6

The Crown Movie Theater collected $705 yesterday. If 54 adults paid $9.50 each and each child paid $3.50 less than each adult, how many children paid admissions?

Three steps, in this order. Adults: 54 × $9.50 = $513. Money left for children: $705 − $513 = $192. Child price: $9.50 − $3.50 = $6.00. Children: $192 ÷ $6 = **32**. The trap is reading "$3.50 less" as the child price itself — $192 ÷ $3.50 is about 55, which is not on the list, so the paper does not even reward that mistake. Check it back: 32 × 6 = 192, and 192 + 513 = 705. ✓

75. (D) $180

Mock 9.25 · Quant 1 · Q21

A coat is on sale for $144 after a discount of 20%. Find the original price.

After a 20% discount you pay 80% of the original price. So 80% of the price = $144, and the full price = 144 ÷ 0.8 = **$180**. Check: 20% of 180 = 36, and 180 − 36 = 144. ✓ The trap is adding 20% of 144 back on: 144 + 28.80 = 172.80 — not a choice here, but a common slip. Rule: working backwards from a sale price, divide by the part you paid.

76. (E) $30.23 Mock 9.25 · Quant 1 · Q22

Rachel worked one Saturday from 8:30 a.m. until 3 p.m. at the rate of $4.65 per hour. How much did she receive?

First the hours: 8:30 a.m. to 12:00 is 3½ hours, and 12:00 to 3:00 is 3 more, so 6½ hours. 6½ × $4.65 = $27.90 + $2.325 = $30.225, which rounds to **$30.23**. Quick check: about 6½ hours at about $5 is about $32, so only (E) is near. Rule: count time across noon in two pieces — up to 12, then after 12.

---

77. (A) 0.40 in ⚠ MISSED TWICE Mock 9.25b · Quant 2 · Q13

If the outer diameter of a metal pipe is 3.01 inches and the inner diameter is 2.21 inches, the thickness of the metal is _____.

The difference 3.01 − 2.21 = 0.80 covers the metal on **both sides** of the pipe (a diameter goes all the way across). So one wall is 0.80 ÷ 2 = **0.40** inches. The trap answer 0.80 isn't here, but that is the step most people forget. Rule: draw a quick picture — a ring has metal on the left and on the right.

---

78. (C) 2:15 p. m. Mock 9.25 · Quant 2 · Q14

Johnny picked apples from 9:00 a. m. to 11:30 a. m. and gathered 200 apples. He wants to pick a total of at least 600 apples before 7:15 p. m. If he plans to pick apples at the same rate, what is the latest time that he can start picking apples again?

His rate: 200 apples in 2½ hours. He still needs 600 − 200 = 400 apples, which is twice as many, so twice the time: 5 hours. Counting back 5 hours from 7:15 p.m. gives **2:15 p.m.** The trap is planning for all 600 apples again (7½ hours), which would give 11:45 a.m. — not even a choice. Rule: subtract what is already done before you work out the time left.

---

79. (C) $14.50 Mock 10.4c · Quant 1 · Q5

A garbage company charges $10 for each of the first 3 months of service and $16 each month after the first 3 months. What is the average cost per month of using this garbage company for the first 12 months?

Total: 3 × $10 = $30, plus 9 × $16 = $144, so $174 for 12 months. $174 ÷ 12 = **$14.50**. The trap is $13.00, the average of $10 and $16 — but there are many more $16 months than $10 months. Rule: average = total ÷ count, not the middle of two prices.

---

80. (B) 0.75 Mock 10.5 · Quant 2 · Q22

Sophie is making bows for packages and uses 1.25 meters of ribbon per bow. If she has 12 meters of ribbon and makes as many complete bows as possible, how much ribbon, in meters, will remain?

12 ÷ 1.25 = 9.6, so she can make 9 complete bows using 9 × 1.25 = 11.25 m. Left over: 12 − 11.25 = **0.75** m. The trap is 0.60, reading the ".6" of 9.6 as meters — it is 0.6 of a bow. Rule: a remainder must be found in the original units.

Day 5 · Questions 81–99

81. (D) 11

Mock 8 · Quant 1 · Q18

Each of the equal legs of an isosceles triangle is 3 more than the base, and the perimeter is 30. Find the measure of one of the equal legs.

Let the base be b; each leg is $b + 3$, and there are two of them. So $b + 2(b + 3) = 30$, which is $3b + 6 = 30$, so $b = 8$ and each leg is **11**. Check: 8 + 11 + 11 = 30. ✓ The trap is 8, the base, and stopping one line early is the most common way to lose this one. Name the *smallest* quantity as your letter, then build the others off it — but circle the one that was asked for.

82. (D) The two lines are neither parallel nor perpendicular

Mock 9 · Quant 1 · Q16

Which of the following statements is true about the two lines below?

Parallel lines never meet — these two cross, so parallel is out at once, and that also kills (A) and (C). Perpendicular means they cross at a **square corner**, 90°; these cross at a clear slant, making two wide angles and two narrow ones. So they are **neither**. Note that (C) can never be true of any pair of lines — parallel and perpendicular exclude each other, so you can cross it out before you even look at the picture.

83. (B) Cone

Mock 9 · Quant 1 · Q17

Which of the following figures is not a polygon?

A polygon is a **flat, closed** shape made only of **straight** sides. Decagon (10 sides), triangle (3), rhombus (4) and pentagon (5) all qualify. A **cone** fails twice over: it is a solid, not a flat figure, and its surface curves. The –gon ending is a useful tell — it means "angle", so anything ending in –gon is a polygon and is counting its corners.

84. (C) 13 cm

Mock 9 · Quant 1 · Q21

The perimeter of a 6-sided polygon is 56 cm. The length of three sides is 9 cm each. The length of two other sides is 8 cm each. What is the length of the missing side?

Count the sides before you add anything: 3 + 2 = 5 of the six are given, so exactly one is missing. Those five come to 3 × 9 = 27 and 2 × 8 = 16, together **43**. The perimeter is the whole way round, so the last side is 56 − 43 = **13 cm**. (E) is tempting because you are not told the shape — but perimeter only cares about the total, never the shape. Whenever a perimeter question gives you all-but-one side, it is a subtraction.

85. (E) m∠1 + m∠3 = 120°

Mock 9.25 · Quant 1 · Q20

NOTE: Figure not drawn to scale.
Which of the following is true?

The three angles of any triangle add to 180°. Angle 2 is 60°, so the other two must share 180 − 60 = **120°**. (C) looks true in the picture, but the figure is *not drawn to scale*, so you cannot trust how it looks. (B) is printed without an ending, so it cannot be the answer. Rule: when a figure is not to scale, use only facts that are always true (like the 180° rule).

86. **(E) 21** Mock 9.25b · Quant 2 · Q24

How many edges does a heptagonal prism have? (A heptagon is a polygon with 7 sides.)

A prism has a top face, a bottom face, and straight edges joining them. Top heptagon: 7 edges. Bottom heptagon: 7 edges. Upright edges joining matching corners: 7. Total: 7 × 3 = **21**. (C) 14 is the trap of counting only the top and bottom — or of counting corners (a heptagonal prism has 14 corners). Rule: any prism with an n-sided base has **3n edges** (a cube: 3 × 4 = 12 ✓).

87. **(A) 9** ⚠ MISSED TWICE Mock 9.25c · Quant 1 · Q12

Triangles *ABE* and *ACD* are similar. Find the length of *DE*.

Similar triangles are the same shape, so their sides grow by the same factor. Side AB = 4 matches the big side AC = 6 + 4 = 10: the big triangle is $^{10}/_{4}$ = 2.5 times bigger. So AD = 2.5 × AE = 2.5 × 6 = 15. DE is AD minus AE: 15 − 6 = **9**. (B) 15 is the trap — that is the whole bottom side AD, not the piece DE. Rule: **find the scale factor from one matching pair of sides, then use it.**

88. **(A) 22** Mock 10.4 · Quant 1 · Q23

Shape *G* is made from pieces *H* and *F*. What is the perimeter of shape *G*?

Note: Figure not drawn to scale.

The hexagon is six *H* triangles, so each outside edge is 2 and its perimeter is 6 × 2 = 12. The bottom part is two *F* triangles, a 3-by-4 rectangle with perimeter 14. Where they touch, one edge of length 2 disappears from each shape: 12 + 14 − 2 − 2 = **22**. The trap is removing the shared edge only once (24). Rule: perimeter is only the outside edges — inside lines never count.

89. **(E) Rectangle** Mock 10.4b · Quant 1 · Q23

In Figure 2, a cube is divided into two pieces by the plane. What is the shape formed by the intersection of the cube and plane?

The flat plane slices straight across the cube from the upper left to the lower right, cutting through four faces. Each face it crosses leaves one straight side, and opposite sides are parallel and meet at right angles, so the cut is a **rectangle**. The trap is (B) Square: the sides that run slanted across the faces are longer than the sides that run straight across. Rule: a slice through two opposite edges of a cube is a rectangle.

90. **(B) 25°** Mock 10.4c · Quant 1 · Q20

In the figure below, $\overline{AB}$ and $\overline{CD}$ are parallel. In addition, $m\angle ABC = 100°$, $m\angle ACD = 130°$, and $\overline{AD}$ bisects $\angle BAC$. What is $m\angle D$?

∠ACD = 130°, so CD makes 180° − 130° = 50° with the line to the right of C. AB is parallel to CD, so ∠BAC = 50° too. AD cuts it in half: ∠DAC = 25°. In triangle ACD: 180° − 130° − 25° = **25°**. The trap is 50° (forgetting the bisector); ∠ABC = 100° is extra information. Rule: parallel lines copy angles, and a bisector halves them.

91. **(B) $5\sqrt{2}$ miles** Mock 10.4c · Quant 1 · Q25

Rafael walks 2 miles east, 2 miles north, 4 miles east, 5 miles north, and 5 miles west. What is the shortest distance between Rafael and his starting point?

Net east: 2 + 4 − 5 = 1 mile. Net north: 2 + 5 = 7 miles. Straight-line distance = $\sqrt{(1^2 + 7^2)} = \sqrt{50} = \mathbf{5\sqrt{2}}$ **miles**. The trap is 18 miles, the total distance walked, not the distance from the start. Rule: combine east–west and north–south separately, then use the Pythagorean theorem.

92. **(D) 4 × 8 × 12** Mock 10.5 · Quant 2 · Q12

The top view of a rectangular package of 6 tightly packed balls is shown. If each ball has a radius of 2 centimeters, which of the following are closest to the dimensions, in centimeters, of the rectangular package?

Each ball is 2 × 2 = 4 cm across. Two balls side by side: 8 cm; three in a row: 12 cm; one ball high: 4 cm. So the box is **4 × 8 × 12**. The trap is 2 × 4 × 6, which uses the radius instead of the diameter. Rule: a ball takes up its diameter in every direction.

93. **(C) 16** Mock 10.5c · Quant 2 · Q8

In triangle *ABC* shown, if *AB* = *BC*, what is the perimeter of the triangle?

AB = *BC* = 5, and *AC* = 6, so the perimeter is 5 + 5 + 6 = **16**. The trap is 11, adding only the two labeled sides. Rule: use the given equal sides to fill in the missing length.

94. **(D) area in square units of a rectangle measuring 2 units by 11 units** Mock 7 · Quant 1 · Q17

The perimeter of a rectangle measuring 3 × 8 units is equal to the

First get the target: perimeter = 2(3 + 8) = **22**. Now test each choice — area 24, pentagon perimeter 15, triangle area 12, **rectangle area 2 × 11 = 22**, square area 484. Only (D) lands on 22. Answer (E) is the trap: 22 is the number you want, not the side length.

95. **(B) 14.0 square units** Mock 7 · Quant 2 · Q24

A triangle has vertices located at the points (3, 2), (3, 9), and (7, 2) on the coordinate plane. Find its area.

Sketch it. Two points share $x = 3$, so that side is vertical and 9 − 2 = **7** long; two share $y = 2$, so that side is horizontal and 7 − 3 = **4** long. It is a right triangle, so area = ½ × 4 × 7 = **14**. Answer (D) 28 forgets the ½ — that is the rectangle, not the triangle.

96. **(A) $\frac{1}{4}$** ⚠ MISSED TWICE Mock 8 · Quant 1 · Q17

Determine the ratio of the area of a circle with radius 2 to the area of a circle with radius 4.

Areas are 4π and 16π, so the ratio is $4\pi/16\pi = \mathbf{\frac{1}{4}}$ — the π cancels, as it always does in a ratio of circle areas. (B) ½ is the ratio of the *radii*, and it is the trap the whole question is built on. The rule worth keeping: **double the radius and the area goes up four times**, because the radius is squared. Triple it and the area is nine times bigger.

97. (E) $8a$ ⚠ MISSED TWICE

Mock 8 · Quant 1 · Q19

If the area of a square is $4a^2$, find its perimeter.

Area of a square is side², so the side is $\sqrt{(4a^2)} = 2a$. Perimeter is four sides: $4 \times 2a = \mathbf{8a}$. Watch the units of the answer: a perimeter is a *length*, so it cannot carry a square on it — that alone rules out (B) and (D) before you calculate anything. Checking whether an answer is a length or an area is one of the cheapest ways to delete two choices.

98. (C) 10.1 kilograms

Mock 8 · Quant 2 · Q17

Metal screws weigh 3 kilograms per cubic meter. The screws are packaged in a cube whose edge is 1.5 meter. Find the weight of the cube. Round the answer to the nearest tenth of a kilogram.

Volume of a cube is edge³: $1.5 \times 1.5 \times 1.5 = 3.375$ cubic metres. At 3 kg per cubic metre that is $3 \times 3.375 = 10.125$ kg. Then do what the last line says — round to the nearest **tenth**: **10.1 kg**. Choice (B) is the unrounded answer, sitting there to catch anyone who stops one step early. When a question tells you how to round, that instruction *is* part of the question.

99. (D) $16 + 6\sqrt{2}$

Mock 8 · Quant 2 · Q22

Find the total area of the figure.

Split the shape into the two pieces you know. The square is $4 \times 4 = \mathbf{16}$. The triangle has base 4 and height $3\sqrt{2}$, so its area is $\frac{1}{2} \times 4 \times 3\sqrt{2} = 2 \times 3\sqrt{2} = \mathbf{6\sqrt{2}}$. Total: $\mathbf{16 + 6\sqrt{2}}$. The two parts cannot be added into one term because one is a plain number and the other carries a root — leaving them side by side is the finished answer, not an unfinished one. A rough check: $\sqrt{2} \approx 1.4$, so the figure is about $16 + 8.5 = 24.5$, and the square alone is 16, which is the right sort of size.

Day 6 · Questions 100–118

100. (C) 374.04

Mock 9 · Quant 1 · Q18

What is the area of the regular hexagon shown below?

A regular hexagon is six identical triangles meeting at the centre. Each has base 12 (a side) and height 10.39 (the marked segment, the *apothem*), so each is $\frac{1}{2} \times 12 \times 10.39 = 62.34$, and six of them make **374.04**. The shortcut is the same sum tidied up: area = ½ × perimeter × apothem = $\frac{1}{2} \times 72 \times 10.39 = 374.04$. Watch the traps: (B) 124.68 is 12×10.39 (one rectangle, no halving), and (D) 748.08 is forgetting the ½ altogether — exactly double the right answer. A doubled or halved answer sitting in the list is almost always a sign the ½ is the point of the question.

101. (C) 216 cm

Mock 9 · Quant 1 · Q19

The area of a given rectangle is 24 centimeters. If the measure of each side is multiplied by 3, what is the area of the new figure?

Try it on a real rectangle: 4 by 6 has area 24. Multiply both sides by 3 and you get 12 by 18, area 216. So the area is **9 times** bigger, not 3 times. That is the rule worth keeping: if every length is multiplied by k, area is multiplied by k^2 (and volume by k^3). (B) 72 is 24×3, the trap for scaling the area once instead of scaling both sides. Picking friendly numbers and testing is faster and safer than arguing about it in your head.

102. (C) $2\sqrt{13}$ Mock 9.25b · Quant 1 · Q14

What is the distance between (−10, −13) and (−16, −9) along the line connecting them?

Make a right triangle. Across: from −10 to −16 is 6. Up/down: from −13 to −9 is 4. The distance is the hypotenuse: $6^2 + 4^2 = 36 + 16 = 52$, and $\sqrt{52} = \sqrt{(4 \times 13)} = \mathbf{2\sqrt{13}}$ (about 7.2). (B) 10 is the trap of just adding 6 + 4 — walking round two sides is longer than the straight line.

103. (D) $24\sqrt{3}$ Mock 9.25b · Quant 1 · Q19

Calculate the area of the hexagon. $OP = 2\sqrt{3}$, $AB = 4$.

The lines from O cut the hexagon into **6 equal triangles**. Look at one: triangle OAB has base AB = 4 and height OP = $2\sqrt{3}$. Its area is $\frac{1}{2} \times 4 \times 2\sqrt{3} = 4\sqrt{3}$. Six of them: $6 \times 4\sqrt{3} = \mathbf{24\sqrt{3}}$. (B) $32\sqrt{3}$ and (A) $96\sqrt{3}$ are too big — check that you used the height OP, not a side, and halved base × height. (E) 24 drops the $\sqrt{3}$. Rule: break an odd shape into pieces you know.

104. (D) $\frac{7}{9}$ ⚠ MISSED TWICE Mock 9.25b · Quant 2 · Q12

From a piece of tin in the shape of a square 6 inches on a side, the largest possible circle is cut out. Of the following, the ratio of the area of the circle to the area of the original square is closest in value to _____.

The largest circle has diameter 6, so radius 3. Circle: $\pi \times 3^2 \approx 3.14 \times 9 \approx 28.3$. Square: $6 \times 6 = 36$. Ratio: $28.3 \div 36 \approx 0.785$ (it is always $\pi/4$). Now turn the choices into decimals: 0.8, 0.67, 0.6, **0.778**, 0.5. The closest is $\frac{7}{9}$. (A) $\frac{4}{5} = 0.8$ is the near-miss trap, off by 0.015 against 0.007.

105. (D) $100 - 25\pi$ cm² Mock 9.25b · Quant 2 · Q21

If the radius of the circle below is 5 cm, what is the area of the shaded region, in terms of π? *(Note: $A = \pi r^2$)*

Shaded = square − circle. The circle touches all four sides, so the square's side equals the diameter, 10 cm: square = $10 \times 10 = 100$. Circle = $\pi \times 5^2 = 25\pi$. Shaded = $\mathbf{100 - 25\pi}$ cm² (about 21.5). (C) uses 5π, which forgets to square the radius. (A) and (B) put the π on the square, which has no curve in it.

106. (A) 10 cm² ⚠ MISSED TWICE Mock 9.25b · Quant 2 · Q22

In the figure below, *JH* is the height of the triangle and measures 8 cm, *GI* = 5 cm, and *GH* = *HI*. What is the area of triangle *JHI*?

GH = HI makes the big triangle isosceles, so its height lands exactly in the middle of the base: JI = 5 ÷ 2 = 2.5 cm. Triangle JHI has base 2.5 and height 8: $\frac{1}{2} \times 2.5 \times 8 = \mathbf{10}$ cm². (C) 20 is the area of the *whole* triangle GHI; (E) 40 forgets the $\frac{1}{2}$.

107. (C) 6 units left, 8 units up Mock 9.25b · Quant 2 · Q25

Which translation moves a point from (4, −3) to (−2, 5)?

Do *x* and *y* separately. *x*: from 4 to −2 is a change of −6, so **6 left**. *y*: from −3 to 5 is a change of +8, so **8 up**. (D) has the right sizes but both directions backwards — that would move (−2, 5) back to (4, −3). Rule: new minus old tells you the direction: negative = left/down, positive = right/up.

108. (D) 12 Mock 9.25c · Quant 1 · Q9

Two cardboard boxes have equal volume. The dimensions of one box are 6 × 6 × 10. If the length of the other box is 3 and the width is 10, what is the height of the second box?

First box: 6 × 6 × 10 = 360. Second box: 3 × 10 × height = 30 × height. So 30 × height = 360, and height = **12**. Check: 3 × 10 × 12 = 360 ✓. (C) 10 is the trap of matching numbers instead of volumes.

109. (C) 8 Mock 10.5c · Quant 2 · Q25

The length of rectangle A is 6 times the length of rectangle B. The width of rectangle A is $\frac{4}{3}$ the width of rectangle B. If the area of rectangle A is x times the area of rectangle B, what is the value of x?

Area is length × width, so the scale factors multiply: 6 × $\frac{4}{3}$ = **8**. Test it: if B is 1 by 3 (area 3), A is 6 by 4 (area 24), and 24 ÷ 3 = 8. The trap is $\frac{9}{2}$, from using $\frac{3}{4}$ instead of $\frac{4}{3}$. Rule: when both sides are scaled, the area is scaled by the product.

110. (A) 2 and 4 Mock 7 · Quant 1 · Q25

Which of the following pairs of numbers does NOT have an average equal to half the average of 8 and 16?

The average of 8 and 16 is 12, and half of that is **6**. Now average each pair: **2 and 4 give 3** — not 6. Every other pair averages exactly 6. On a NOT question, work out the target number first, then hunt for the one that misses it.

111. (B) $520 Mock 7 · Quant 2 · Q9

The tablet model that has the highest current price was originally introduced at a price of

Two steps, in order. First find the tallest **dark** bar: that is *Q*, at about $150. Then read *Q*'s light bar: **$520**. Answer (A) $620 is what you get by finding the tallest bar on the whole graph instead of the tallest current price.

112. (E) 119.2 Mock 8 · Quant 1 · Q13

The table represents the frequency distribution of students' weights. Find the mean weight and round to the nearest tenth.

A frequency table is not a plain list: each weight has to be counted as many times as its frequency, which is what the empty $p_i f_i$ column is inviting you to fill in. 98×2 = 196, 105×1 = 105, 109×1 = 109, 120×3 = 360, 126×2 = 252, 132×4 = 528, total **1,550**. The number of students is 2+1+1+3+2+4 = **13**. 1,550 ÷ 13 = 119.23… → **119.2**. The classic error is dividing by 6 (the number of rows), which gives about 258 and is not even on the list. *Divide by the number of people, not the number of rows.*

113. (B) 306 ⚠ MISSED TWICE Mock 8 · Quant 1 · Q2

The car dealership sold 1,800 cars last year. According to the chart, how many Mazdas did the dealership sell?

Mazda is 17% of the 1,800 cars, so you want 0.17 × 1,800. Split it: 10% is 180, 7% is 126, and 180 + 126 = **306**. Three of the other choices are other slices of the same pie — 36% is 648, 29% is 522, 18% is 324 — so reading the wrong row of the chart is the whole trap. Point at the label before you multiply.

114. (C) 16% Mock 8 · Quant 1 · Q21

In 2002, what percent of all television shows were detective stories?

Read the question's year first, then find the right corner of the table. The 2002 column totals **50** shows and the Detective row shows **8** of them, so 8/50. Multiply top and bottom by 2 to get 16/100 = **16%**. The 250 in the corner is the grand total for all four years and is not wanted here — 8/250 would give about 3%. The habit that saves you: before you divide, say out loud what the bottom number is a total *of*.

115. (D) 12% Mock 8 · Quant 1 · Q22

The total number of cartoons in the year 2004 and 2005 represent what percent of all shows in the years 2002 through and including 2005?

Top of the fraction: cartoons in 2004 and 2005 only, 14 + 16 = **30**. Bottom: *all* shows across all four years, the corner cell, **250**. 30/250 = 3/25 = 12/100 = **12%**. This question is the mirror image of the one before it, and it is testing whether you noticed the switch: there the total was one year, here it is four. (B) 24% comes from dividing by 125. **Match the bottom of the fraction to the words "of all…" in the question.**

116. (D) $c - a < b + c$ Mock 8 · Quant 2 · Q18

Examine the graph and then select the correct answer.

Read the four values off the graph first and write them down: $a = 10$, $b = 30$, $c = 90$, $d = 50$. Then it is arithmetic. (A) 50 > 130? No. (B) 50 = 130? No. (C) 50 = 40? No. (D) 90 − 10 = 80 and 30 + 90 = 120, so 80 < 120 ✓. (E) 10 + 90 = 100 and 30 + 50 = 80, so 100 < 80 is false. Good habit: **extract the numbers from the picture once**, then work in numbers only.

117. (E) $y = 40x + 300$ ⚠ MISSED TWICE Mock 9 · Quant 1 · Q13

Karen gets paid a weekly salary and a commission for every sale that she makes. The table below shows the number of sales and her pay for different weeks. Which of the following equations represents Karen's weekly pay?

The commission is the *change* in pay per extra sale. From 2 sales to 7 sales the pay goes from \$380 to \$580: that is \$200 for 5 sales, so **\$40 a sale**. The salary is what is left when she sells nothing: at 2 sales the commission is 2 × 40 = \$80, so the salary is 380 − 80 = **\$300**. Hence $y = 40x + 300$. Test it on a row you did not use: 8 sales → 40(8) + 300 = \$620 ✓. (A) $y = 90x + 200$ fits the first row (90 × 2 + 200 = 380) but fails the others — that is what happens when you check only one row. In a table, slope needs **two** rows.

118. (A) Mean ⚠ MISSED TWICE Mock 9 · Quant 1 · Q23

Which measure for the center of a small sample set is most affected by outliers?

Test it. Take 2, 3, 4, 5: the mean is 3.5 and the median is 3.5. Now swap the 5 for 100: the mean jumps to 27.25 while the median only moves to 3.5. The **mean** uses every value in the sum, so one wild number drags it; the median only cares about the middle position, and the mode about what repeats. Rule: **one extreme value moves the mean and barely touches the median** — which is why house prices and incomes are usually reported as medians.

Day 7 · Questions 119–137

119. (E) 100

Mock 9 · Quant 2 · Q16

Dwayne has received the following scores on his math tests: 78, 92, 83, 97. What score must Dwayne get on his next math test to have an overall average of at least 90?

Work backwards from the total he needs. Five tests averaging 90 means a total of 5 × 90 = **450**. He already has 78 + 92 + 83 + 97 = **350**. So the next test must be 450 − 350 = **100**. (B) 98 and (D) 99 are near-misses from a slip in the addition, and they are meant to feel plausible because 100 looks like an extreme answer. Rule worth keeping: for any "what do I need to average…" question, turn the average into a total first — averages are hard to steer, totals are just subtraction.

120. (C) $ 800

Mock 9.25b · Quant 1 · Q17

Mike bought 10 shares of Zooko stock at the closing price on Tuesday and sold them at the closing on Friday. How much money did Mike lose on his investment?

Read the two points you need, and only those. Tuesday: $120 a share. Friday: $40 a share. Loss per share: 120 − 40 = $80. He had 10 shares: 80 × 10 = **$800**. (A) $80 is the trap — that is the loss on just *one* share. Rule: after reading a graph, go back and check what the question multiplies by.

121. (E) $\frac{1}{216}$

Mock 9.25b · Quant 2 · Q23

A number cube with the faces numbered 1 through 6 is rolled three times. What is the probability that all three rolls will result in the number one?

Each roll gives a 1 with probability $\frac{1}{6}$. For three separate rolls to *all* happen, multiply: $\frac{1}{6} \times \frac{1}{6} \times \frac{1}{6} = \mathbf{\frac{1}{216}}$. (C) $\frac{3}{216}$ is the trap of adding the three chances instead of multiplying. Rule: **"and" in probability means multiply** — the chance gets smaller.

122. (D) 45

Mock 9.25c · Quant 1 · Q1

The average of three numbers is 5. What is three times the sum of the three numbers?

Average × how many = sum: 5 × 3 = 15. The question wants **three times** that sum: 3 × 15 = **45**. (B) 15 is the trap of stopping one step early. Rule: read the last line of the question again before you choose.

123. (C) $600

Mock 9.25c · Quant 1 · Q25

Josie bought 10 shares of Zariche stock at the closing price Monday and sold them at the closing price on Friday. What was Josie's profit on this investment?

Read the two points you need. Monday: $180 a share. Friday: $240 a share. Profit per share: 240 − 180 = $60. Ten shares: 60 × 10 = **$600**. (A) $60 is the trap — that is the profit on just one share. Don't be distracted by the big dip on Tuesday: only the buy day and the sell day matter.

124. (C) 25 Mock 9.25c · Quant 2 · Q5

If a Set *R* contains four positive integers whose average is 7, what is the greatest possible value of one of the integers?

Four numbers averaging 7 add up to 4 × 7 = 28. To make one as big as possible, make the other three as **small** as possible. The smallest positive integer is 1, so 1 + 1 + 1 = 3, leaving 28 − 3 = **25**. (E) 36 and (D) 33 are more than the whole total of 28, so they are impossible. Rule: to make one thing as big as possible, squeeze everything else down.

125. (D) $125 Mock 10.4 · Quant 1 · Q15

The graph below shows the balance of a savings account, which was opened in January, over a 6 month period.

The savings account balance in June was how much greater than in January?

Read the two points: June is $175 and January is $50, so the difference is 175 − 50 = **$125**. The trap is $175 — that is the June balance itself, not how much greater it is. Rule: "how much greater" always means subtract.

126. (B) 300 miles Mock 10.4 · Quant 2 · Q13

The graph below shows the distance travelled by a migrating swallow to nesting grounds 800 miles away.

At the beginning of day 2, the swallow had travelled 200 miles. What total distance did the swallow travel from the beginning of day 2 through the end of day 4?

The graph shows *total* miles. At the end of day 4 the total is 500, and at the beginning of day 2 it was 200, so the swallow flew 500 − 200 = **300 miles**. The trap is 500, the running total, not the distance in that stretch. Rule: on a running-total graph, distance over a period = later value − earlier value.

127. (C) 10 Mock 10.5 · Quant 2 · Q15

30, −15, 10, 35, −15, −30, 20
What is the median of the list above?

Put the numbers in order: −30, −15, −15, 10, 20, 30, 35. With 7 numbers, the median is the 4th: **10**. The trap is 35, the middle of the list *before* sorting (or −15, which is the mode). Rule: always sort before you find the median.

128. (E) $ba \geq ab$ Mock 8 · Quant 1 · Q23

If a is a negative integer and b is a positive one, which of the following statements is true?

Multiplication does not care about order, so ab and ba are the same number. That kills (A) at once — a number is never less than itself — and it makes (E) true, because **every number is greater than or equal to itself**. Of the rest: a square is never negative, so (C) and (D) are out; and $a + b > b - a$ simplifies to $a > -a$, which needs a to be positive — it is not. On a MUST-be-true question, the safest-looking statement is very often the answer, and "≥" is about as safe as it gets.

129. (B) $ab \leq 0$ Mock 8 · Quant 2 · Q9

If a and b are integers and $a > 0$ and $b < 0$, select the statement that is true for all cases of a and b.

Positive times negative is always negative, so $ab < 0$ — which certainly makes $\boldsymbol{ab \leq 0}$ true in every case, and makes (A) false in every case. For the rest, one counter-example each is enough: with $a = 1$ and $b = -5$, (C) says $1 > 25$, false; (D) says $1 < -125$, false; (E) says $2 \leq -5$, false. That is the method for any *true for all cases* question: you only need **one** pair of numbers to knock a choice out, so go hunting for an awkward pair — a big negative and a small positive is usually the one that breaks things.

130. (A) ○ Mock 9 · Quant 1 · Q15

What is the 42nd item in the pattern:

Find the length of one full repeat: ▲ ○ ○ □ — **4 items**. Then divide and look at the *remainder*: 42 ÷ 4 = 10 remainder **2**. Ten whole repeats use up 40 items, so the 42nd is the **2nd item of the next repeat**, which is a circle ○. A remainder of 0 would have meant the *last* item of a repeat, not the first — that is where most pattern questions are lost. Say it out loud: "remainder 2 means second in the block."

131. (E) $\frac{1}{12}$ ⚠ MISSED TWICE Mock 9 · Quant 1 · Q25

What is the probability of randomly picking the winner and runner-up from a race of 4 horses and distinguishing which is the winner?

Count the ways the top two can come out. Any of the **4** horses can win, and then any of the **3** that are left can come second: 4 × 3 = **12** different outcomes, and only one of them is your pick — so $\boldsymbol{\frac{1}{12}}$. (C) $\frac{1}{16}$ is 4 × 4, which would let the same horse finish first and second. Because the question says **distinguishing which is the winner**, order matters, and the second pick always has one fewer horse to choose from.

132. (B) 4 ⚠ MISSED TWICE Mock 9.25 · Quant 1 · Q24

How many socks would you need to remove from a drawer containing 10 blue socks, 15 black socks, and 5 red socks, to insure that you have at least 2 matching socks?

Think about the unluckiest pulls. There are only 3 colours, so the first 3 socks could be one of each colour — no pair yet. The 4th sock must match one of them. So **4**. (D) 17 is the trap of thinking about the biggest pile; the numbers of each colour do not matter here. Rule: for "make sure you get a pair", take one of every kind, then add 1.

133. (E) 1, 2, and 3 Mock 9.25b · Quant 1 · Q22

If 6 is a factor of a certain number, what must also be factors of that number?

If 6 goes into the number, then everything that goes into 6 does too: **1, 2 and 3** (and 6). Try the number 6 itself: 1, 2 and 3 all divide it, but 5 does not — so 5 does not *have* to be a factor. That rules out (A) and (D). (B) says "only" and leaves out 1. Rule: on a **must** question, test a simple example and throw out anything it breaks.

134. (A) + 3 Mock 9.25c · Quant 2 · Q24

In the fraction $\frac{1}{\Delta - 3}$, Δ can be replaced by all of the following EXCEPT _____.

A fraction can never have 0 on the bottom. $\Delta - 3 = 0$ when $\Delta =$ **3**, so 3 is the one number that is not allowed. Every other choice gives a bottom that is not zero (−1, −3, −4, −5). Rule: **you can never divide by zero.**

135. (D) 53 Mock 10.4 · Quant 1 · Q4

Pictures for a magazine layout are numbered consecutively starting with 10. They are arranged in numerical order, three to a page, from left to right. Which of the following could be the number of a picture in the middle of a page?

Page 1 holds 10, 11, 12; page 2 holds 13, 14, 15; so the middle pictures are 11, 14, 17, … — they go up by 3, and each leaves remainder 2 when divided by 3. $53 = 3 \times 17 + 2$, so **53** works. The trap: 40, 43, 46 and 64 all leave remainder 1, which makes them the *left* picture on their page. Rule: for repeating groups, sort numbers by their remainder.

136. (A) 4 Mock 10.4b · Quant 1 · Q6

On a certain obstacle course involving 20 participants, 12 scored more than 50 points and 8 scored at least 52 points. How many participants scored exactly 51 points?

Points are whole numbers, so "more than 50" means 51 or higher. Those 12 people include the 8 who scored 52 or more, so $12 - 8 =$ **4** scored exactly 51. The trap is (E): it feels like missing information, but the two groups fit inside each other. Rule: when one group sits inside another, subtract to find the people in between.

137. (C) 23 Mock 10.4b · Quant 1 · Q7

Tricia shelves consecutively numbered books at the library starting with 1. They are placed 8 to a shelf in numerical order. Which of the following numbers would NOT be the third book on a shelf?

Third books are 3, 11, 19, 27, … — each one is 3 more than a multiple of 8. Check: 19 = 16 + 3, 67 = 64 + 3, 83 = 80 + 3, but **23** = 16 + 7, so it is the 7th book on its shelf. Rule: in a repeating group of 8, the remainder after dividing by 8 tells you the position.